Лабораторная работа 2

Структуры данных

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Цель работы

Основная цель работы — изучить несколько структур данных, реализованных в Julia, научиться применять их и операции над ними для решения задач.

Задача со множествами

```
[6]: A = Set([0, 3, 4, 9])
     B = Set([1, 3, 4, 7])
     C = Set([0,1,2,4,7,8,9])

     P = union(intersect(A, B), intersect(A,B), intersect(A,C), intersect(A,C))

     print(P)

     Set([0, 4, 9, 3])
```

Примеры операций со множествами

```
*[11]: X = Set(["a", "b", "c"])
       Y = Set(["a", 1, "d"])
       Z = Set([4, 1, "b", "a"])

[11]: Set{Any} with 5 elements:
        "c"
        "b"
        "a"
        1
        "d"

[12]: union(X, Y)

[12]: Set{Any} with 5 elements:
        "c"
        "b"
        "a"
        1
        "d"

[13]: intersect(X, Z)

[13]: Set{Any} with 2 elements:
        "b"
        "a"
```

Создание массивов

```
[22]: N = 25
      a1 = collect(1:N)
      print(a1)
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25]
```

Создание массивов

```
[24]: a2 = collect(N:-1:1)
      print(a2)
```

```
[25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
```

Создание массивов

```
[30]: tmp = [4,6,3]

[30]: 3-element Vector{Int64}:
       4
       6
       3
```

Создание вектора

```
[44]: using Statistics

      x = 3:0.1:6
      y = exp.(x) .* cos.(x)
      println(mean(y))

      53.11374594642971
```

Массив из квадратов

```
[11]: squares = [i^2 for i in 1:100]
      print(squares)
```

```
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400, 441, 484, 529, 576, 625, 676, 729, 784, 841, 900, 961, 1024, 1089, 1156, 1225, 1296, 1369, 1444, 1521, 1600, 1681, 1764, 1849, 1936, 2025, 2116, 2209, 2304, 2401, 2500, 2601, 2704, 2809, 2916, 3025,
3136, 3249, 3364, 3481, 3600, 3721, 3844, 3969, 4096, 4225, 4356, 4489, 4624, 4761, 4900, 5041, 5184, 5329, 5476, 5625, 5776, 5929, 6084, 6241, 6400, 6561, 6724, 6889, 7056, 7225, 7396, 7569, 7744, 7921, 8100, 8281, 8464, 8649, 8836, 9025, 9216, 9409, 9604, 9801, 10000]
```

Массив простых чисел

```
[89]: using Primes

      myprimes = primes(168)

[89]: 39-element Vector{Int64}:
        2
        3
        5
        7
       11
       13
       17
       19
       23
       29
       31
       37
       41
        ⋮
      107
      109
      113
      127
      131
      137
      139
      149
      151
      157
      163
      167
```